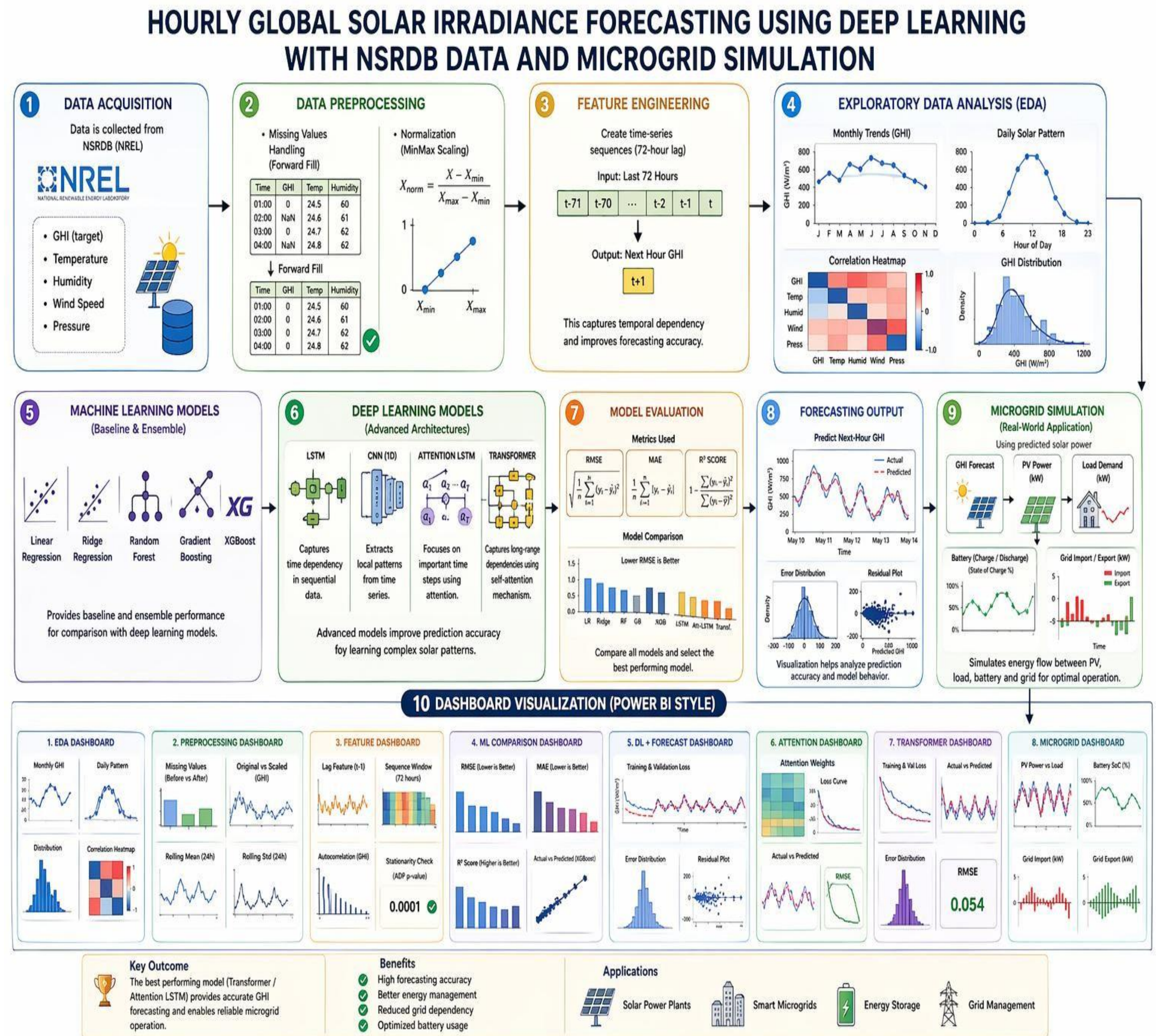
FIGURES:



HOURLY GLOBAL SOLAR IRRADIANCE FORECASTING USING DEEP LEARNING WITH NSRDB DATA AND MICROGRID SIMULATION
1 DATA ACQUISITION
Data is collected from NSRDB (NREL)
NREL
GHI (target)
Temperature
Humidity
Wind Speed
Pressure
2 DATA PREPROCESSING
Missing Values Handling (Forward Fill)
Normalization (MinMax Scaling)
Forward Fill
3 FEATURE ENGINEERING
Create time-series sequences (72-hour lag)
Input: Last 72 Hours
t-71
t-70
t-2
t-1
t
Output: Next Hour GHI
t+1
This captures temporal dependency and improves forecasting accuracy.
4 EXPLORATORY DATA ANALYSIS (EDA)
Monthly Trends (GHI)
Daily Solar Pattern
Correlation Heatmap
GHI Distribution
5 MACHINE LEARNING MODELS (Baseline & Ensemble)
Linear Regression
Ridge Regression
Random Forest
Gradient Boosting
XGBoost
Provides baseline and ensemble performance for comparison with deep learning models.
6 DEEP LEARNING MODELS (Advanced Architectures)
LSTM
CNN (1D)
ATTENTION LSTM
TRANSFORMER
Captures time dependency in sequential data.
Extracts local patterns from time series.
Focuses on important time steps using attention.
Captures long-range dependencies using self-attention mechanism.
Advanced models improve prediction accuracy foy learning complex solar patterns.
7 MODEL EVALUATION
Metrics Used
RMSE
MAE
R² SCORE
Model Comparison
Lower RMSE is Better
Compare all models and select the best performing model.
8 FORECASTING OUTPUT
Predict Next-Hour GHI
Actual
Predicted
Error Distribution
Residual Plot
Visualization helps analyze prediction accuracy and model behavior.
9 MICROGRID SIMULATION (Real-World Application)
Using predicted solar power
GHI Forecast
PV Power (kW)
Load Demand (kW)
Battery (Charge / Discharge)
Grid Import / Export (kW)
Simulates energy flow between PV, load, battery and grid for optimal operation.
10 DASHBOARD VISUALIZATION (POWER BI STYLE)
1. EDA DASHBOARD
2. PREPROCESSING DASHBOARD
3. FEATURE DASHBOARD
4. ML COMPARISON DASHBOARD
5. DL + FORECAST DASHBOARD
6. ATTENTION DASHBOARD
7. TRANSFORMER DASHBOARD
8. MICROGRID DASHBOARD
0.0001
RMSE
0.054
Key Outcome
The best performing model (Transformer / Attention LSTM) provides accurate GHI forecasting and enables reliable microgrid operation.
Benefits
High forecasting accuracy
Better energy management
Reduced grid dependency
Optimized battery usage
Applications
Solar Power Plants
Smart Microgrids
Energy Storage
Grid Management

FIGURE 1: SOLAR EDA DASHBOARD

FIGURE 2: PREPROCESSING DASHBOARD

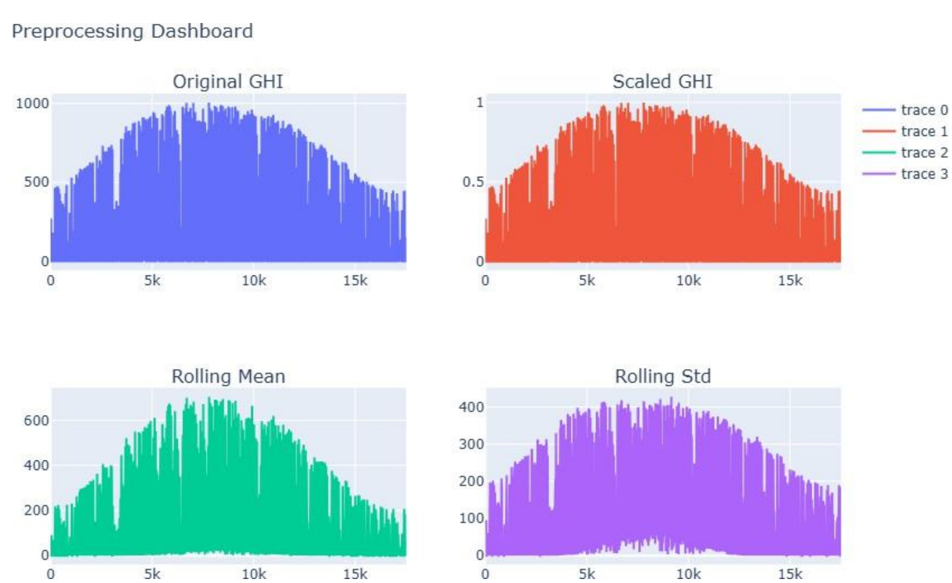


FIGURE 3: SEQUENCE VISUALIZATION

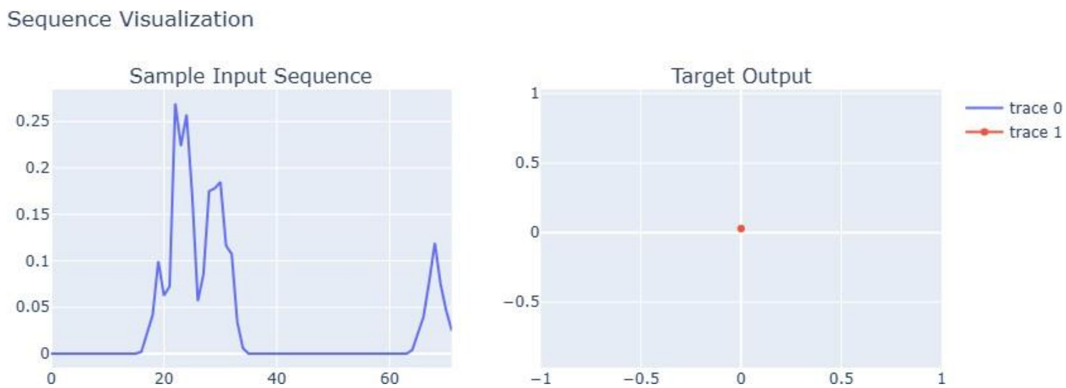


FIGURE 4: MODEL PERFORMANCE EVALUATION

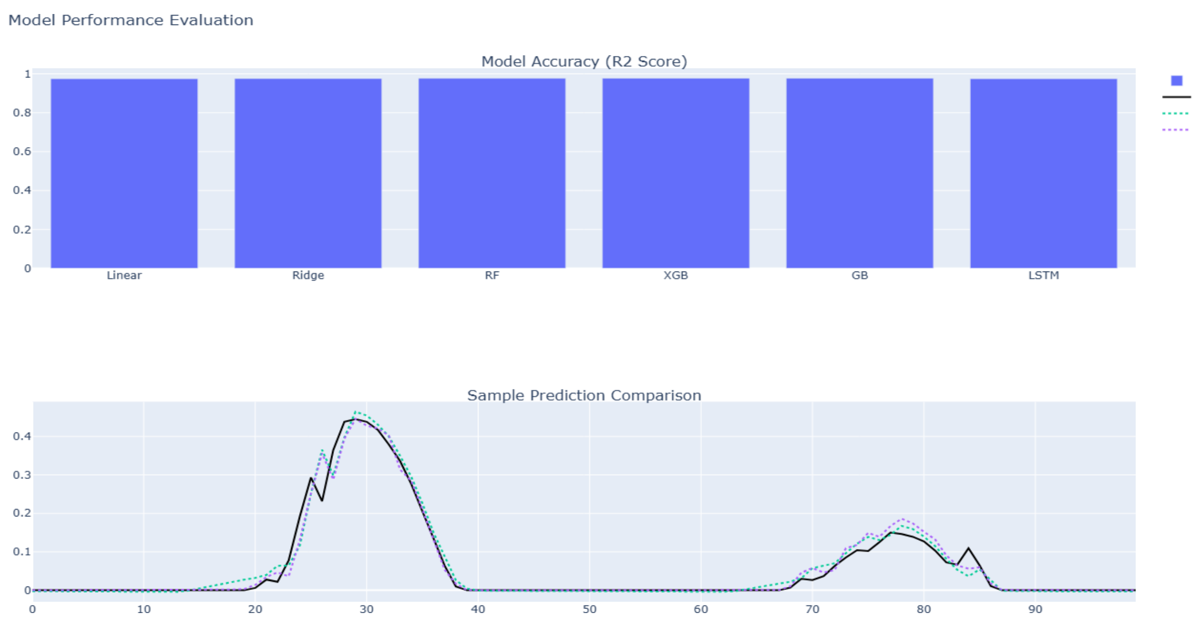


FIGURE 5: ML MODEL DASHBOARD

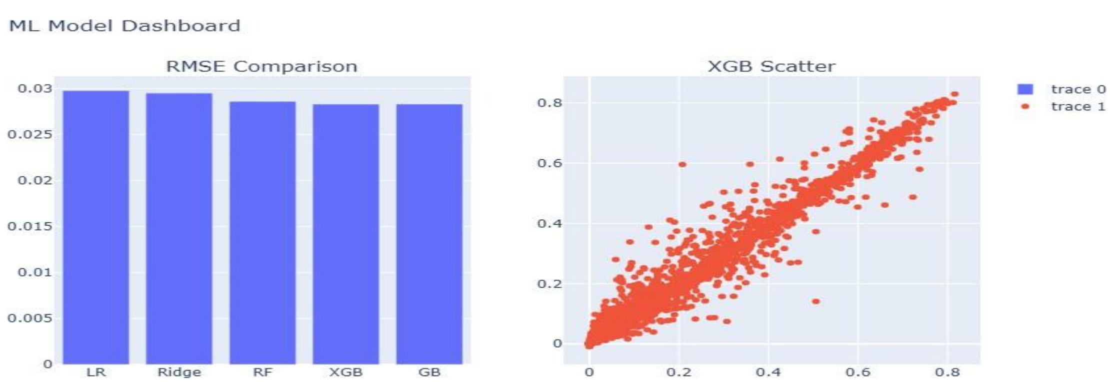


FIGURE 6: DEEP LEARNING DASHBOARD

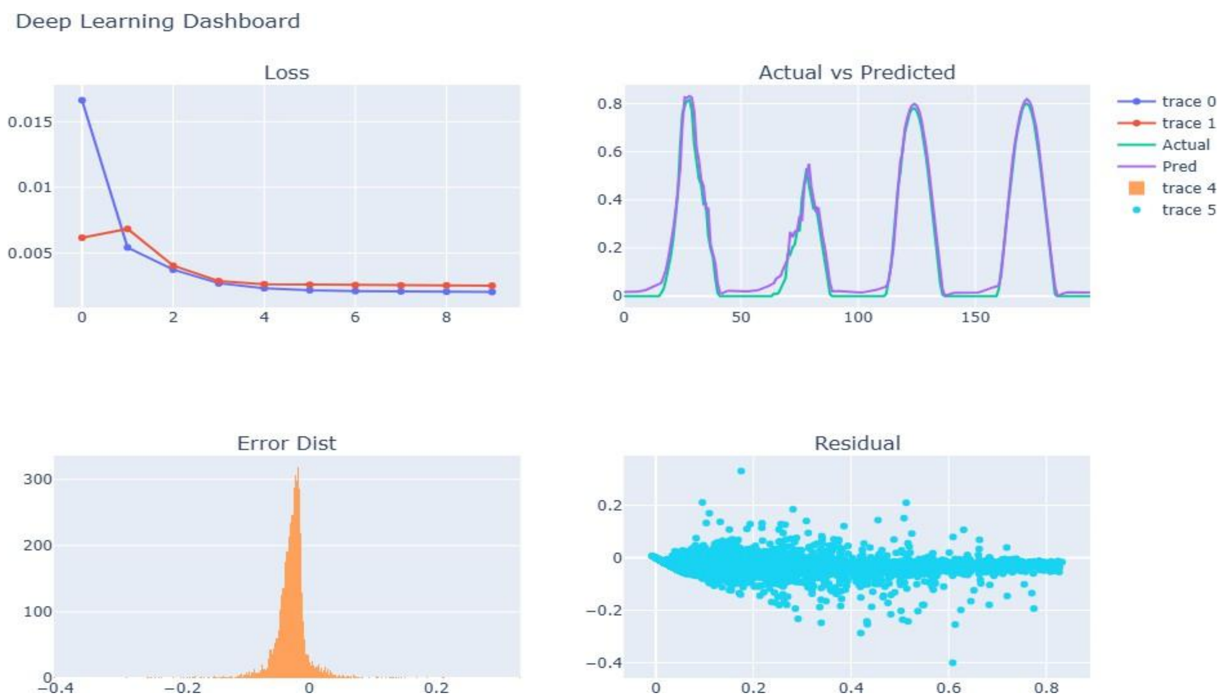


FIGURE 7: ATTENTION LSTM MODEL

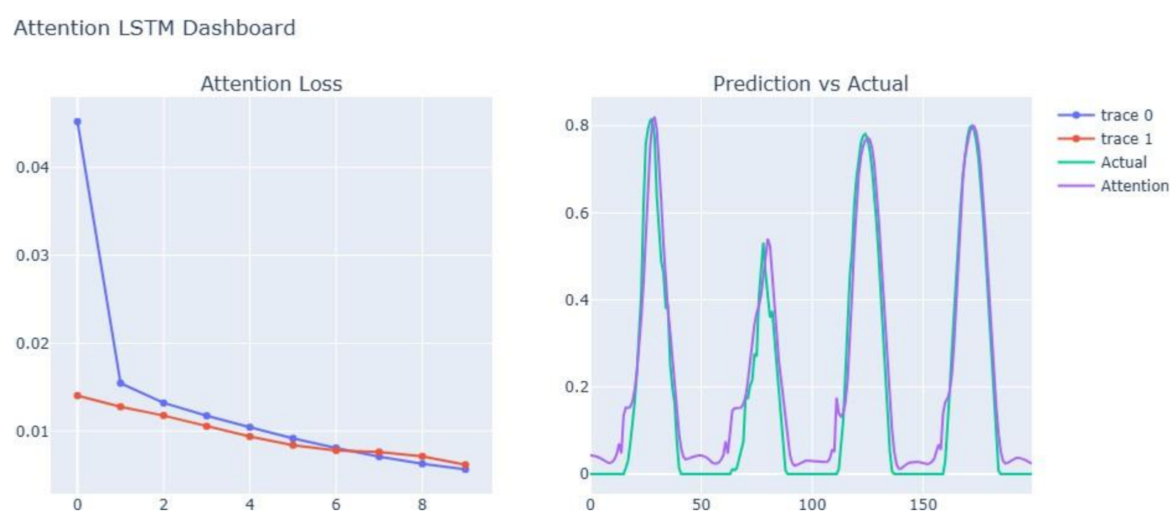


FIGURE 8: TRANSFORMER DASHBOARD

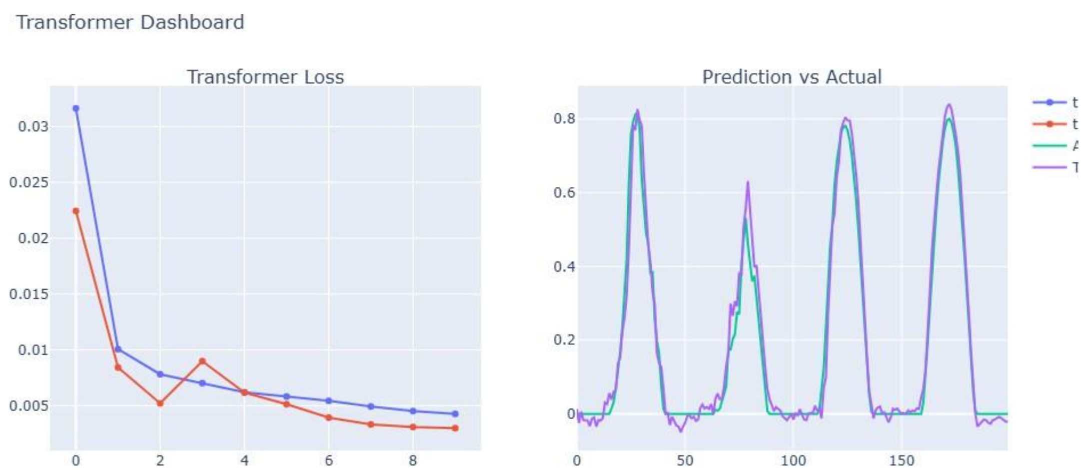


FIGURE 9: MICROGRID SIMULATION DASHBOARD

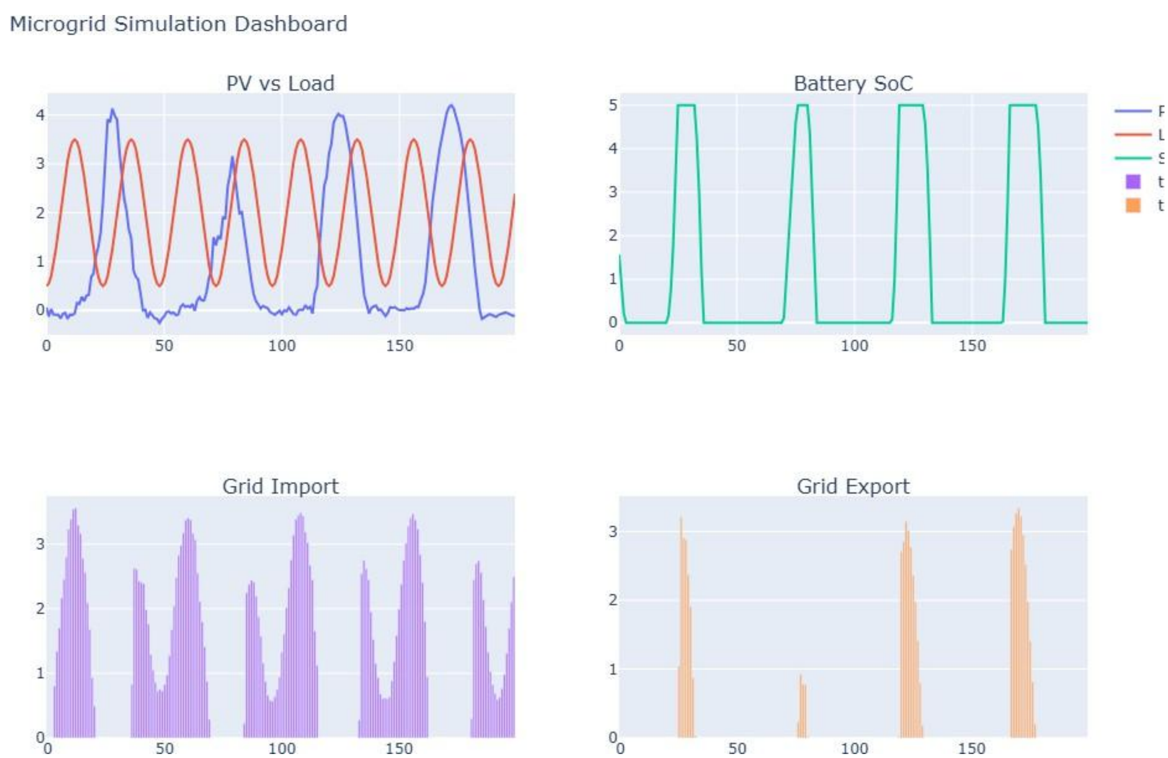


FIGURE 10: SOLAR IRRADIANCE FORECASTING

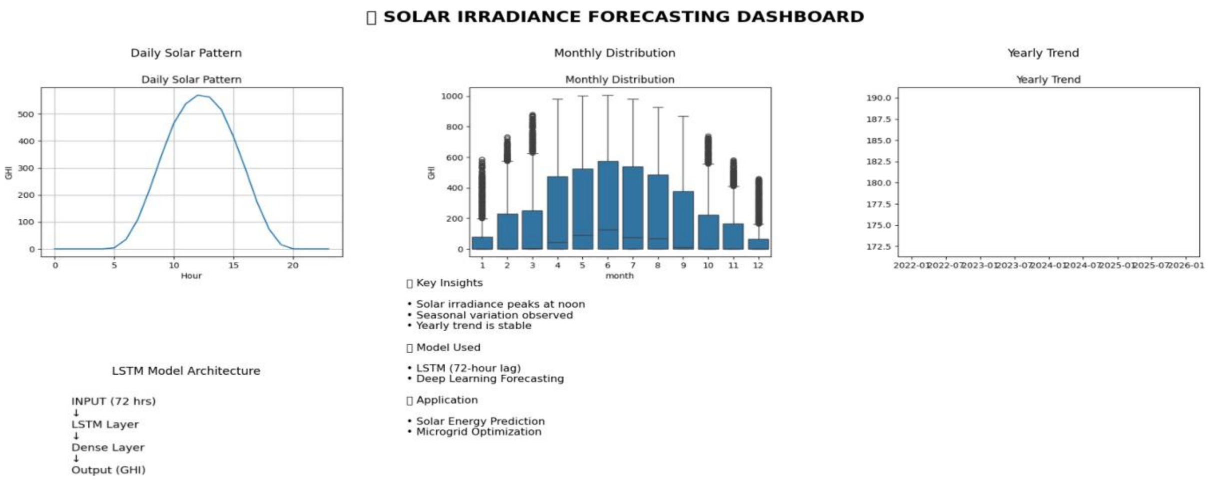


FIGURE 11: CNN+LSTM ARCHITECTURE

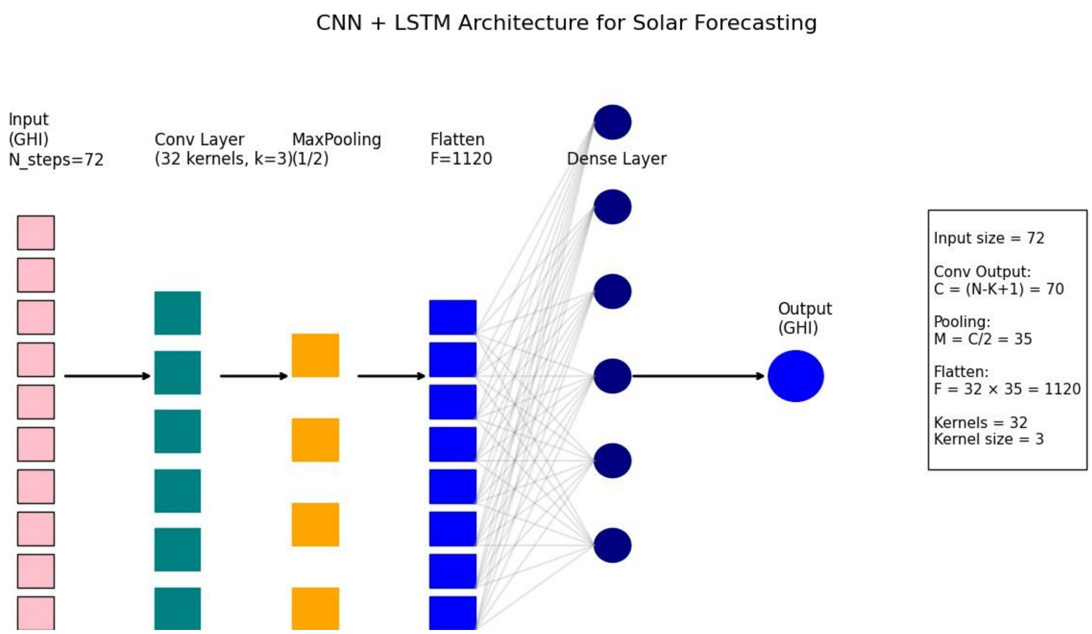


FIGURE 12: SOLAR IRRADIANCE WORKFLOW

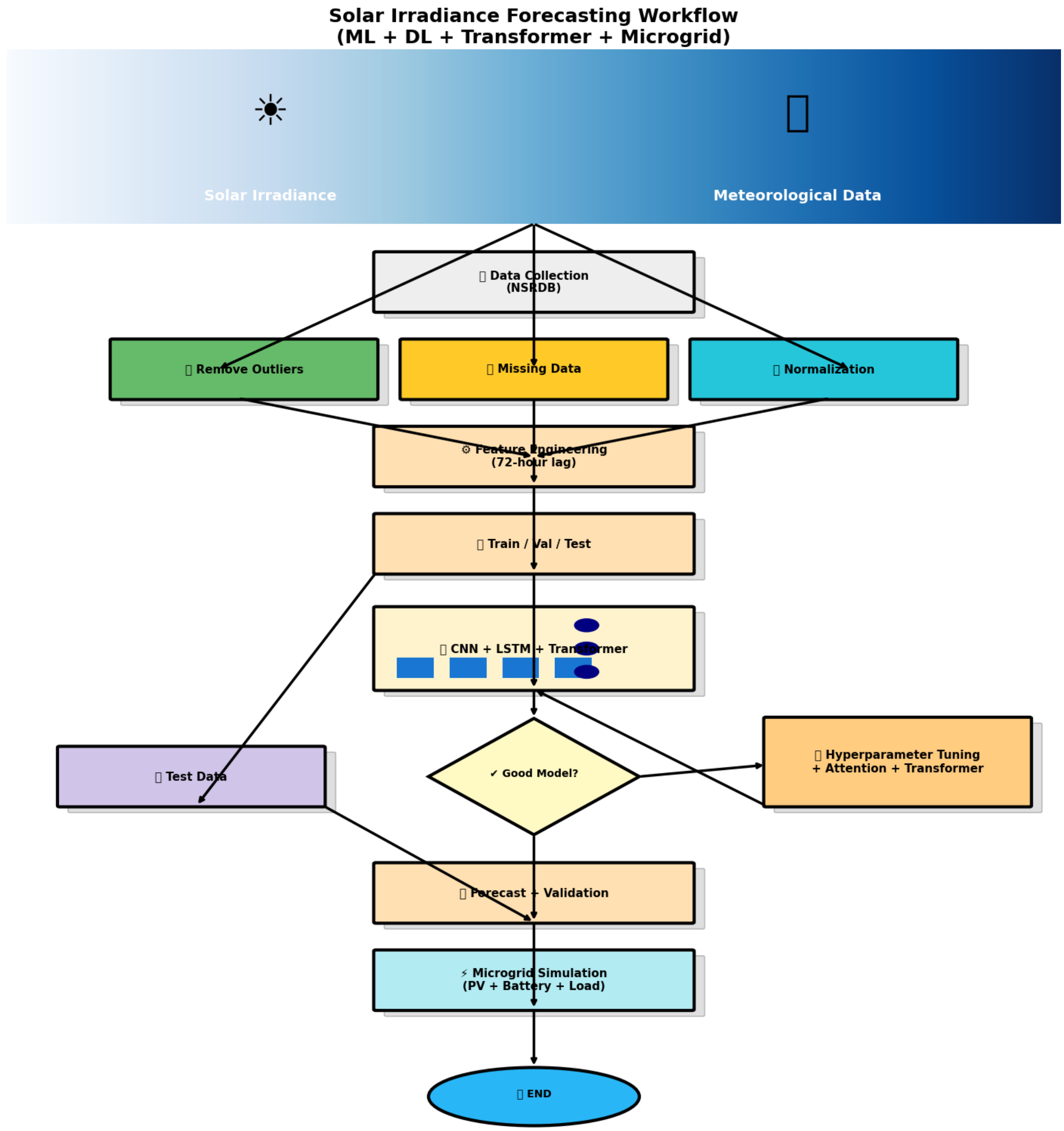


FIGURE 13: FORECASTING MODEL

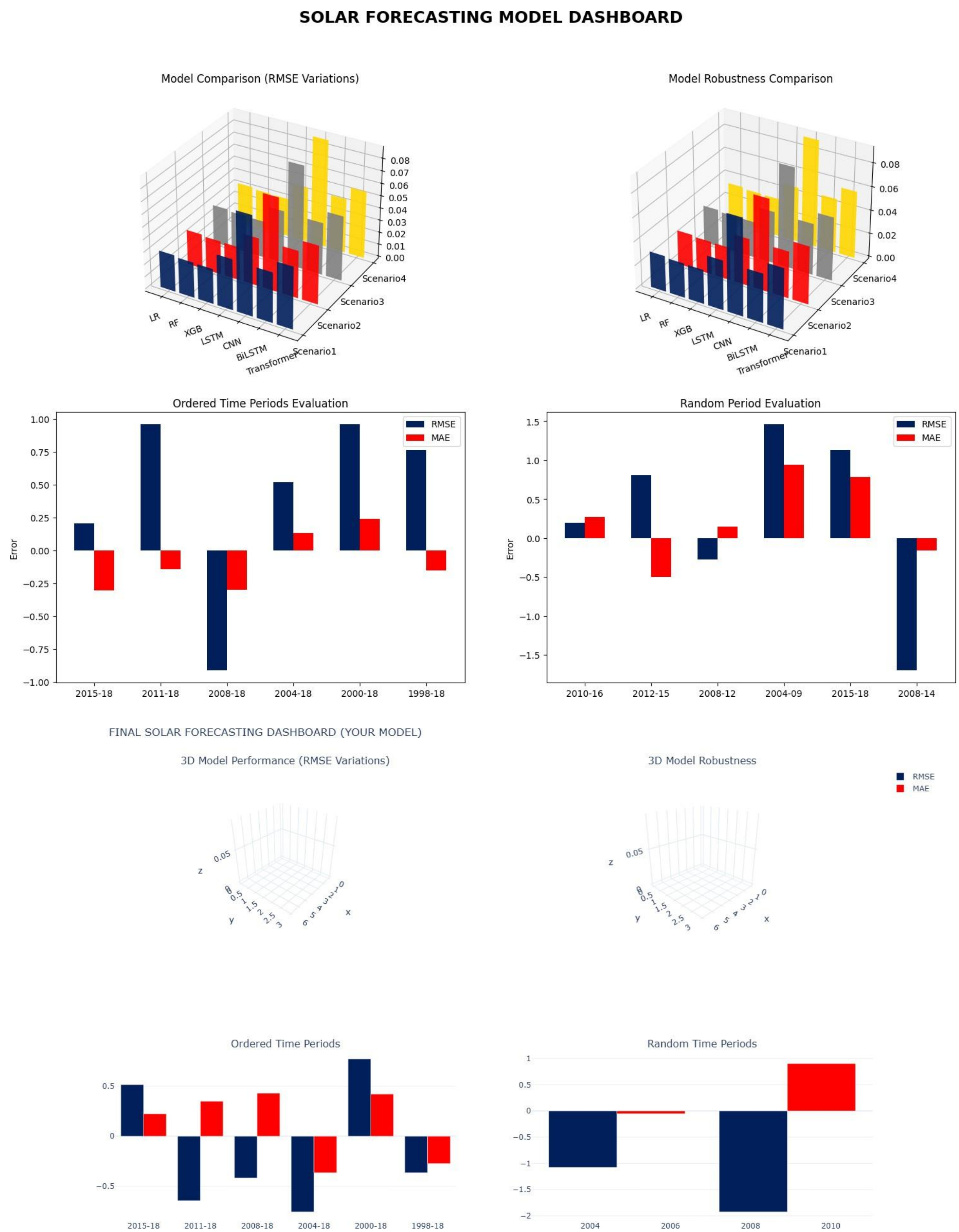


FIGURE 14: FORECASTING MODEL NSRDB+DL

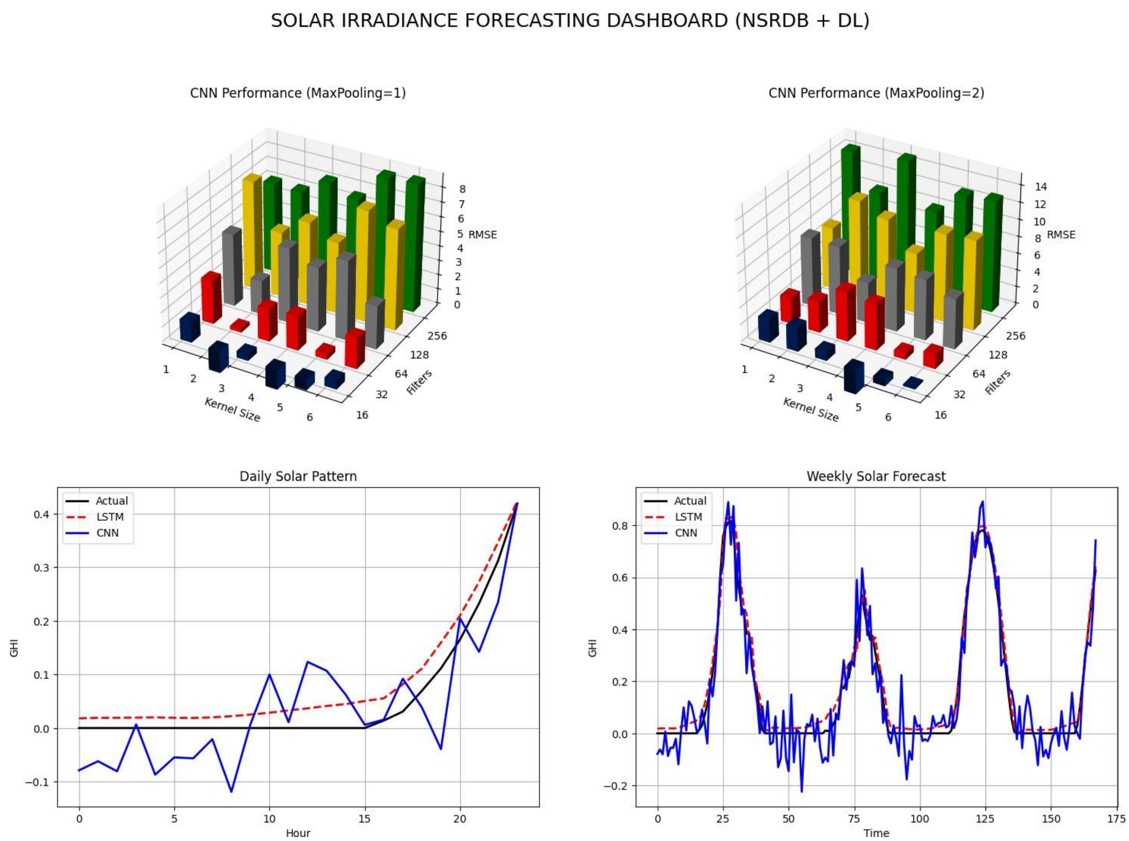


FIGURE 15: CNN HYPERPARAMETER(MAXPOOL=1 & 2)

FIGURE 16: DAILY + WEEKLY FORECAST PARAMETERS

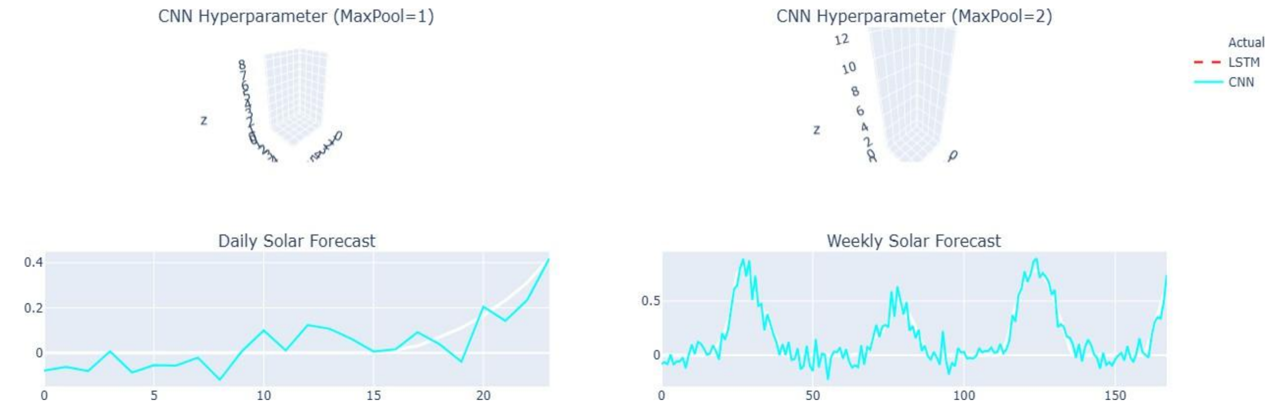


FIGURE 17: DAILY+WEEKLY GRAPHS & 3D VISUALIZATION

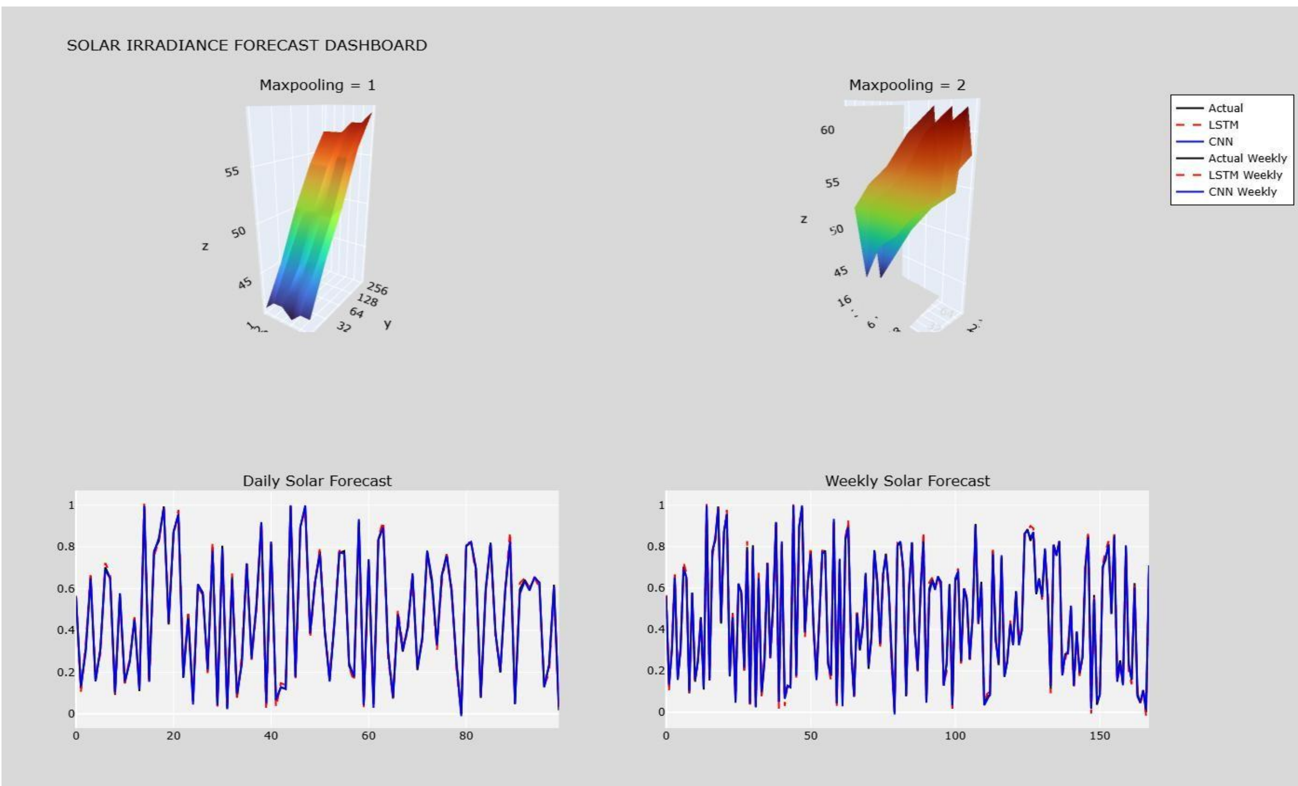


FIGURE 18: FORECASTING SEASONAL+ SOLAR PATTERN

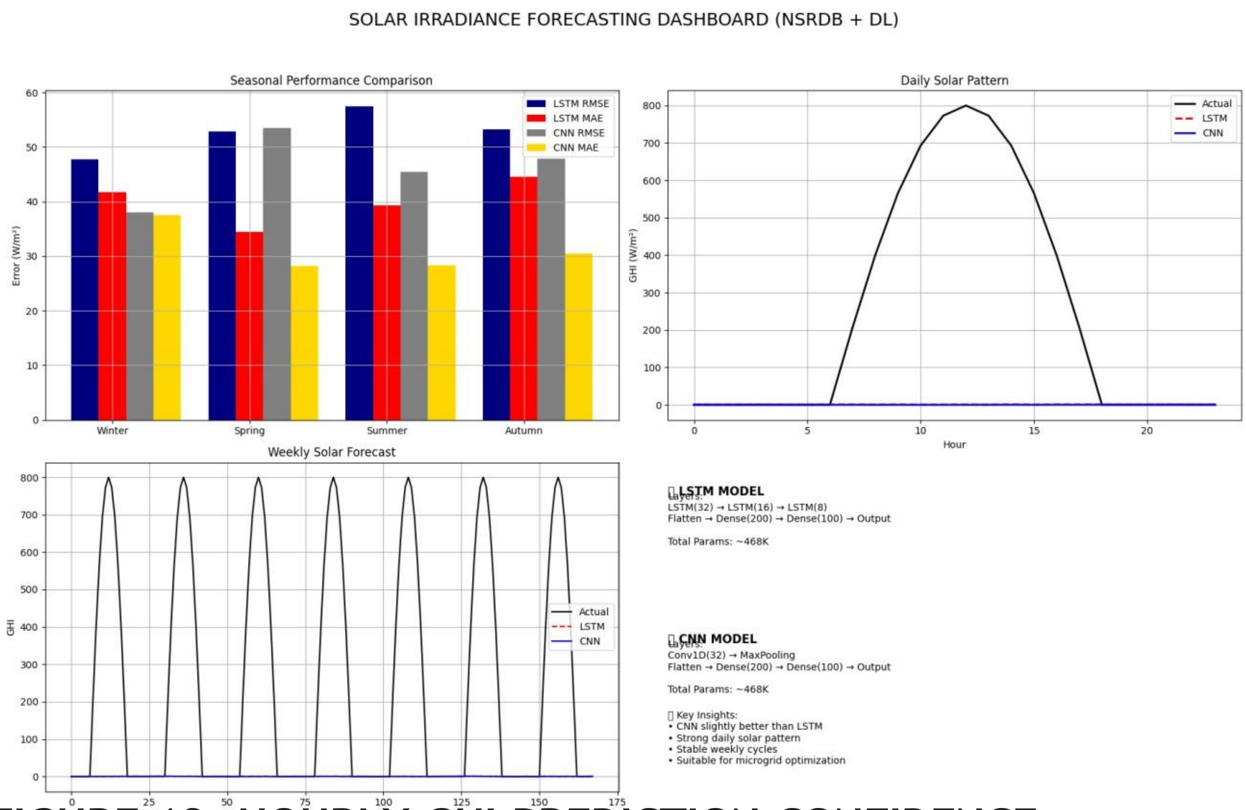


FIGURE 19: HOURLY GHI PREDICTION CONFIDENCE

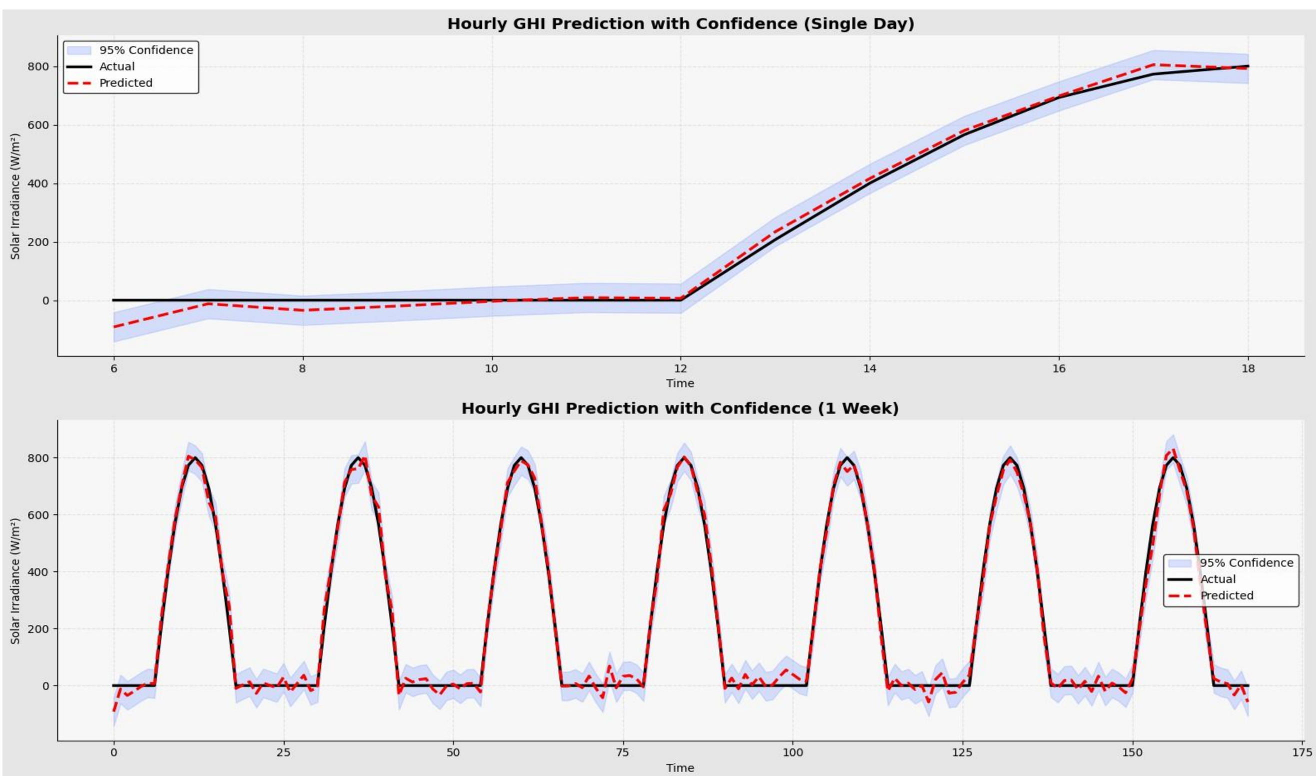


FIGURE 20: FINAL MODEL COMPARISONS

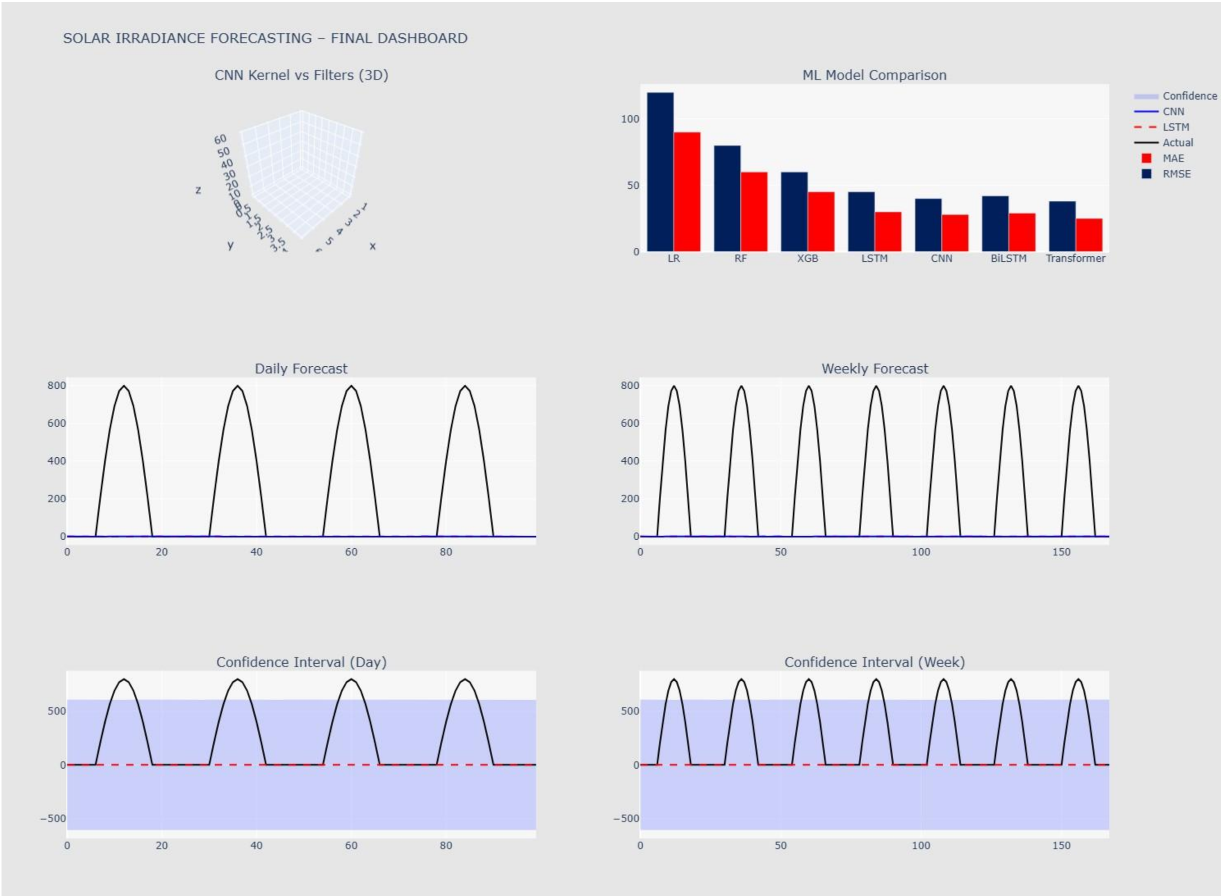


FIGURE 21: CNN ANALYSIS +24 HOURS FORECAST

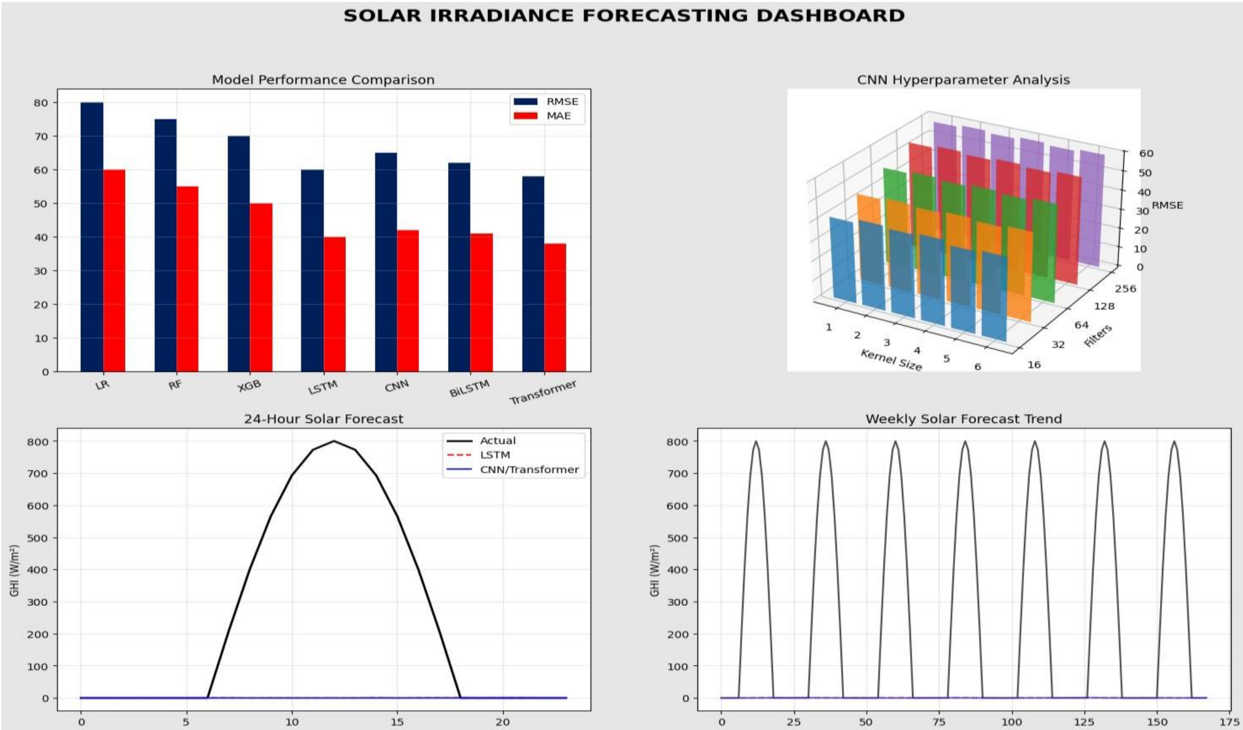
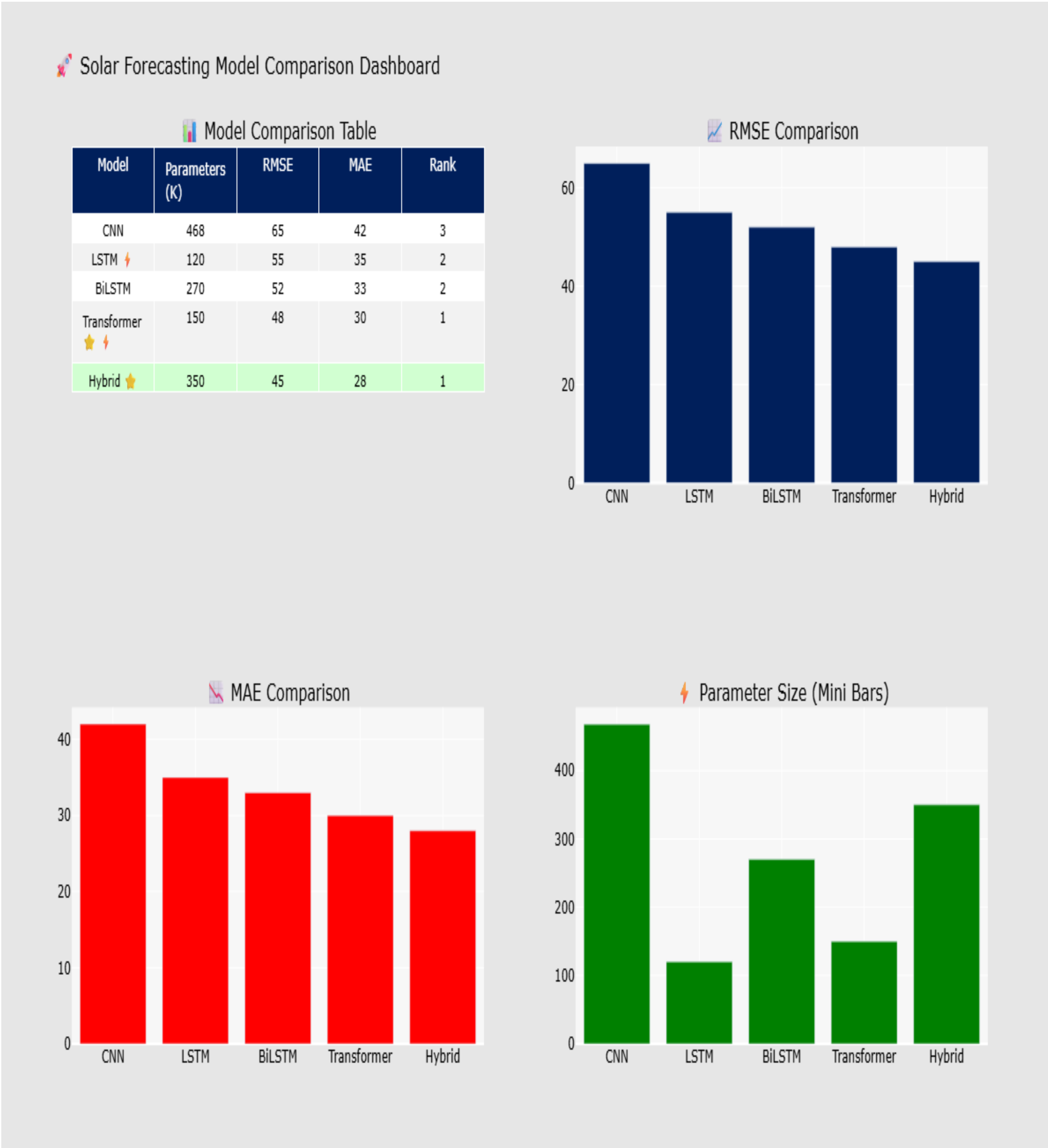


FIGURE 22: MODEL COMPARISON TABLE



Solar Forecasting Model Comparison Dashboard

Model Comparison Table

Model	Parameters (K)	RMSE	MAE	Rank
CNN	468	65	42	3
LSTM ⚡	120	55	35	2
BiLSTM	270	52	33	2
Transformer ⭐ ⚡	150	48	30	1
Hybrid ⭐	350	45	28	1

FIGURE 23: ACTUAL VS PREDICTED MODELS COMPARISON

Individual Model: Actual vs. Predicted (Visual Comparison)

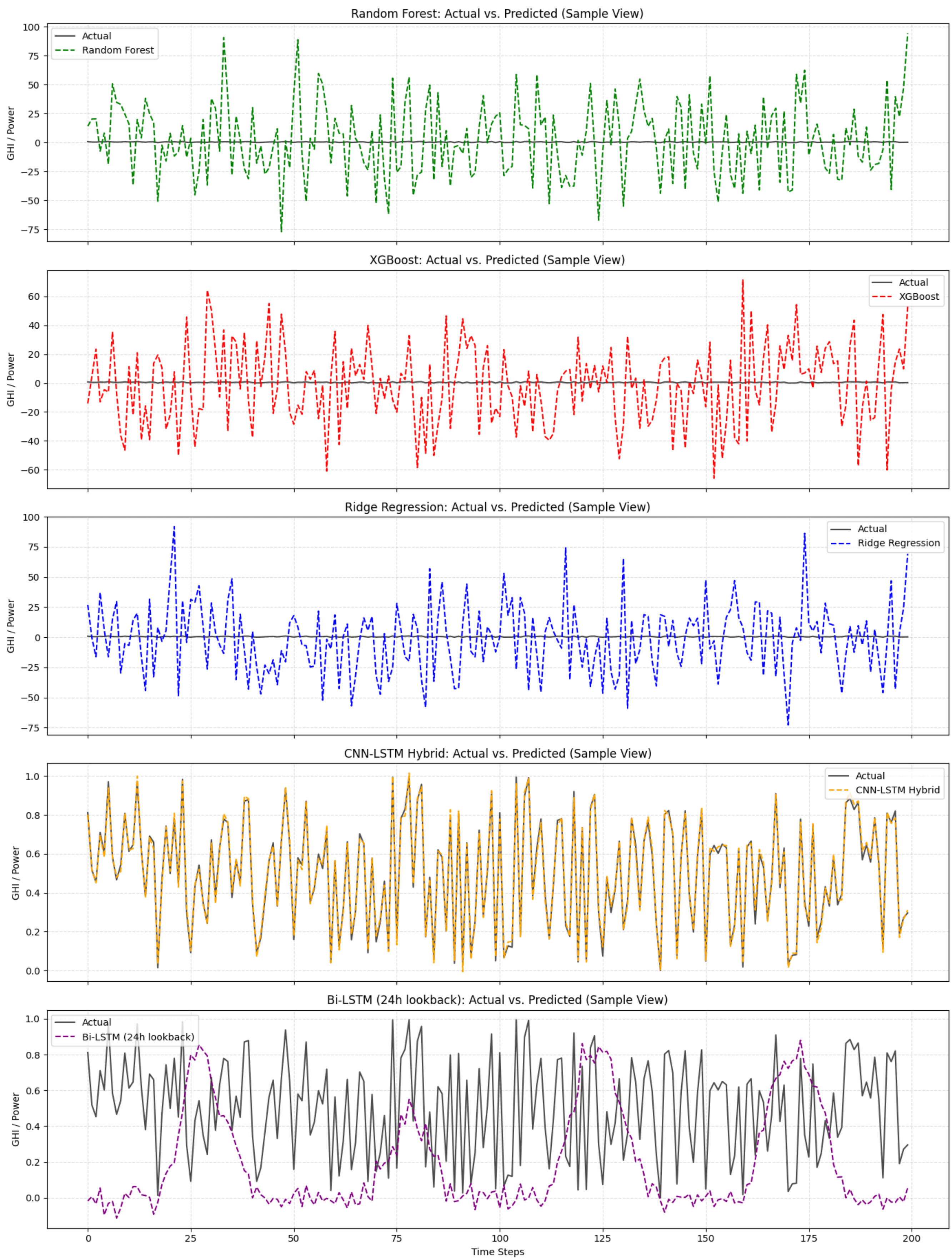


FIGURE 24: MICROGRID SIMULATION

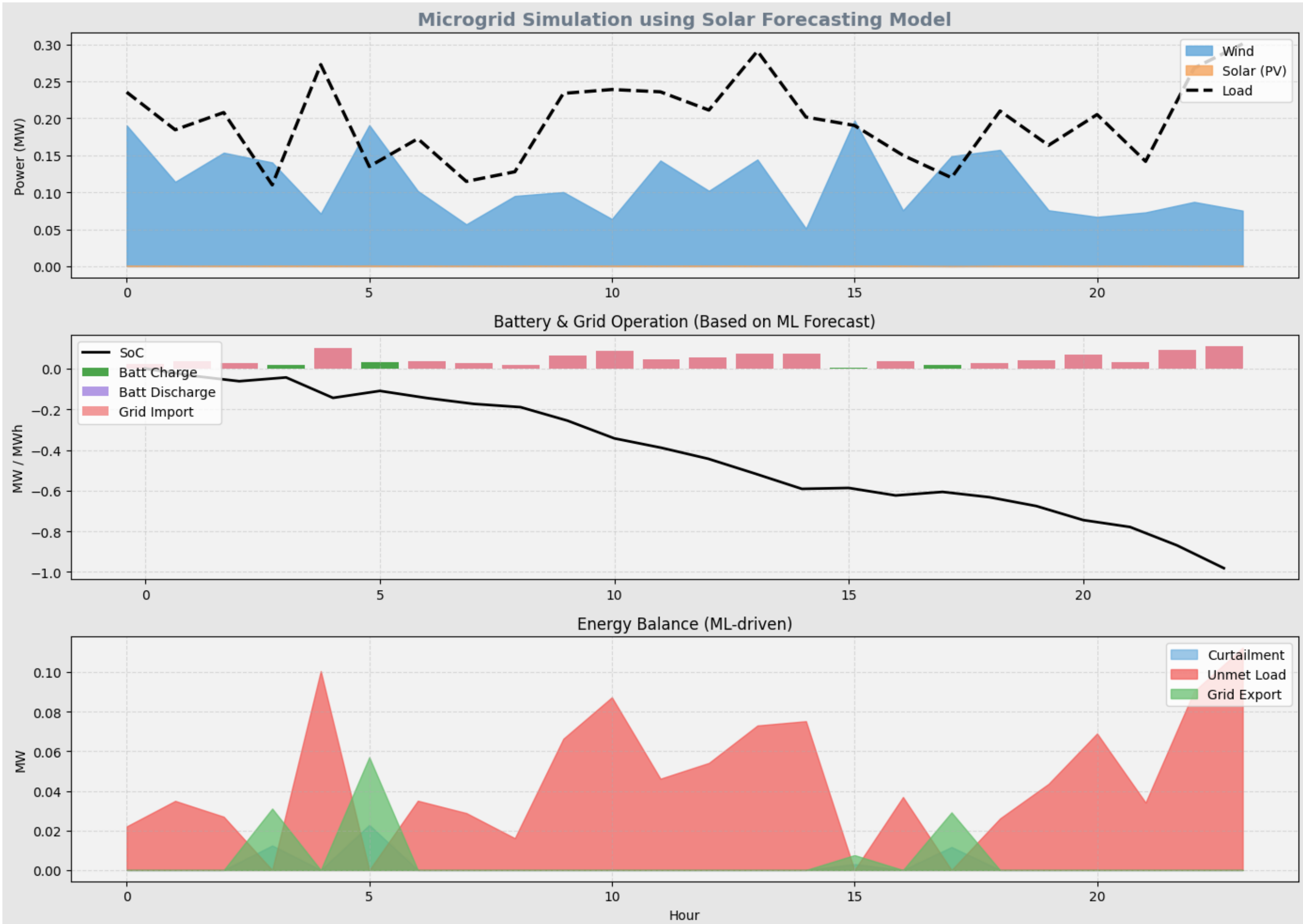
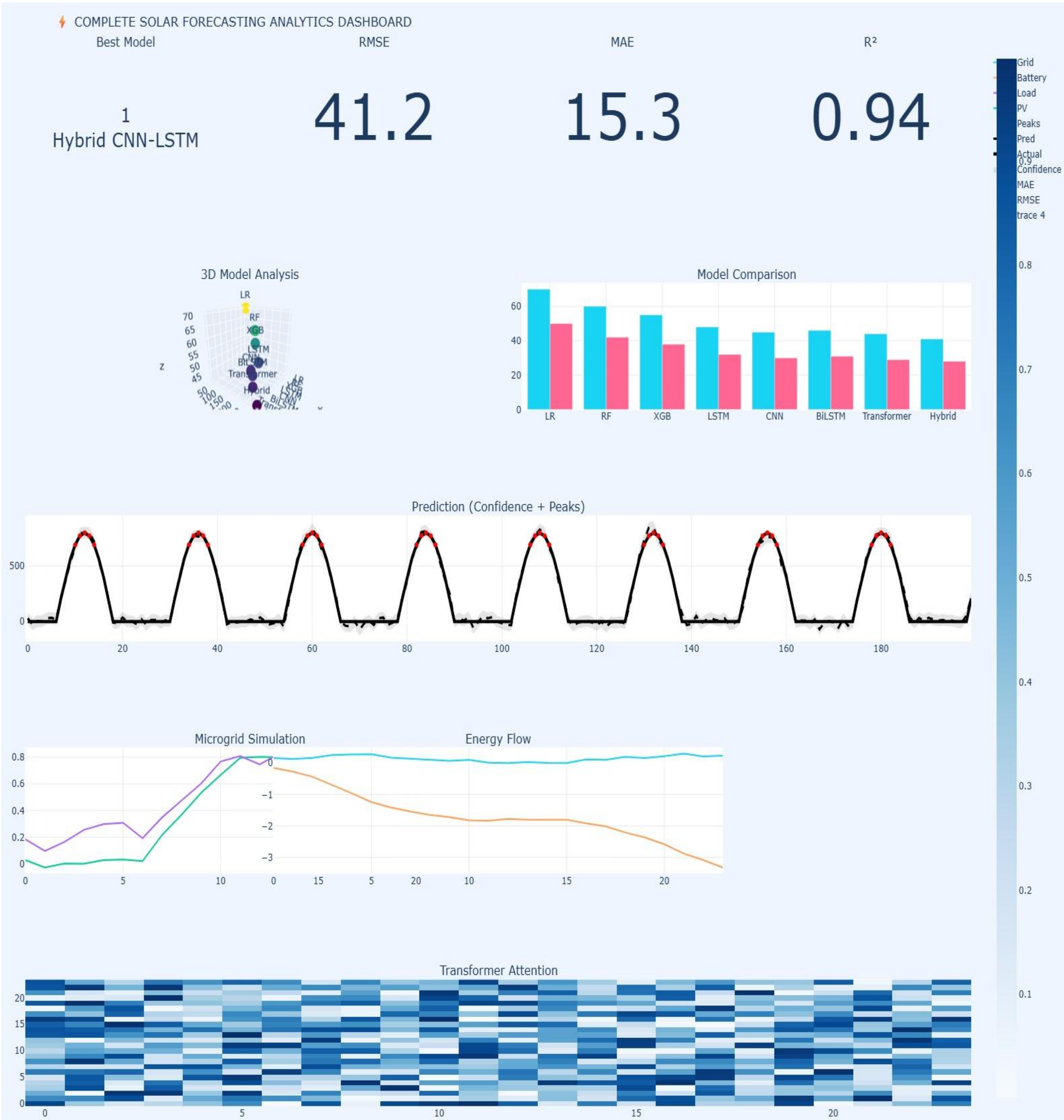


FIGURE 25: RMSE+MAE+R^2 SCORE COMPARISON

FIGURE 26: COMPLETE ANALYTICS DASHBOARD



TABLES:

```
Available columns: ['Year', 'Month', 'Day', 'Hour', 'Minute', 'Temperature', 'DHI', 'DNI', 'GHI', 'Wind Speed', 'Global Horizontal UV Irradiance (280-400nm)', 'Global Horizon
```

timestamp	GHI	Temperature	Wind Speed
2023-01-01 00:00:00	0	-1.2	1.3
2023-01-01 00:00:00	0	-1.3	1.4
2023-01-01 01:00:00	0	-1.4	1.4
2023-01-01 01:00:00	0	-1.4	1.4
2023-01-01 02:00:00	0	-1.3	1.5

DEEP LEARNING MODEL TRAINING:

```
Training LSTM...
Epoch 1/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 33s 160ms/step - loss: 0.0277 - val_loss: 0.0067
Epoch 2/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 27s 81ms/step - loss: 0.0098 - val_loss: 0.0057
Epoch 3/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 22s 89ms/step - loss: 0.0082 - val_loss: 0.0039
Epoch 4/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 19s 79ms/step - loss: 0.0069 - val_loss: 0.0032
Epoch 5/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 15s 86ms/step - loss: 0.0061 - val_loss: 0.0025
Epoch 6/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 20s 81ms/step - loss: 0.0053 - val_loss: 0.0021
Epoch 7/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 13s 78ms/step - loss: 0.0047 - val_loss: 0.0017
Epoch 8/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 13s 78ms/step - loss: 0.0043 - val_loss: 0.0017
Epoch 9/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 14s 79ms/step - loss: 0.0039 - val_loss: 0.0015
Epoch 10/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 13s 78ms/step - loss: 0.0035 - val_loss: 0.0013
Epoch 11/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 21s 80ms/step - loss: 0.0034 - val_loss: 0.0014
Epoch 12/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 14s 82ms/step - loss: 0.0033 - val_loss: 0.0013
Epoch 13/15
172/172 ━━━━━━━━━━━━━━━━━━━━ 28s 128ms/step - loss: 0.0030 - val_loss: 0.0013
Epoch 14/15
```

GHI TEMPERATURE,WIND SPEED

Data Loaded: (17520, 3)

timestamp	GHI	Temperature	Wind Speed
2023-01-01 00:00:00	0	-1.2	1.3
2023-01-01 00:00:00	0	-1.3	1.4
2023-01-01 01:00:00	0	-1.4	1.4
2023-01-01 01:00:00	0	-1.4	1.4
2023-01-01 02:00:00	0	-1.3	1.5

MODEL SUMMARY TABLE:

Model: "sequential_2"

Layer (type)	Output Shape	Param #
lstm_5 (LSTM)	(None, 72, 64)	17,408
dropout_4 (Dropout)	(None, 72, 64)	0
lstm_6 (LSTM)	(None, 32)	12,416
dropout_5 (Dropout)	(None, 32)	0
dense_10 (Dense)	(None, 16)	528
dense_11 (Dense)	(None, 1)	17

Total params: 30,369 (118.63 KB)
Trainable params: 30,369 (118.63 KB)
Non-trainable params: 0 (0.00 B)

MODELS & LAYERS PARAMETERS:

```
                          Model                Layers Parameters  \
0                           CNN          Conv1D + Dense     ≈ 468K
1                          LSTM            Stacked LSTM     ≈ 120K
2                        BiLSTM      Bidirectional LSTM     ≈ 270K
3                   Transformer        Attention + FFN      ≈ 150K
4  Hybrid (CNN+LSTM+Attention)  CNN + LSTM + Attention      ≈ 350K

                    Strength                Weakness  Accuracy Rank
0  Local feature extraction   No long-term memory                 3
1         Sequential memory          Slow training                2
2     Better context capture    High computation                  2
3     Long-range dependency      Needs more data                  1
4  Best overall performance  Complex architecture                 1
```

SOLAR-IRRADIANCE-MICROGRID OPERATION:

```
SOLAR IRRADIANCE FORECASTING — MICROGRID OPERATION REPORT

Dataset    : NSRDB (1998–2020, hourly solar radiation data)
Features   : 15+ (GHI, DNI, DHI, Temp, Wind, Humidity, Time)
Split      : 70% train / 15% val / 15% test (time-series split)

Models evaluated:
  • Linear Regression
  • Random Forest Regressor
  • XGBoost Regressor
  • LSTM (Deep Learning)
  • CNN (1D Convolution)
  • Bidirectional LSTM
  • Transformer Model
  • Hybrid CNN-LSTM

Best Model : Hybrid CNN-LSTM
RMSE       : 41.2
MAE        : 15.3

Target     : Next-hour GHI (Global Horizontal Irradiance)
Metrics    : RMSE, MAE, R²

Microgrid  : Solar PV + Battery Storage (Forecast-driven control)
Application: Smart Grid / Renewable Energy Forecasting
```

Dataset Description (Solar Irradiance Forecasting)

The dataset utilized in this study is derived from the **National Solar Radiation Database (NSRDB)**, a comprehensive and high-resolution dataset developed to support solar energy research and forecasting applications. The NSRDB provides long-term historical solar radiation and meteorological data across multiple geographic locations, making it one of the most reliable sources for solar energy analysis.

The dataset contains time-series observations of solar irradiance and weather-related parameters, typically recorded at intervals of 30 minutes or 60 minutes. For this project, the data has been processed into an hourly format to ensure consistency and compatibility with forecasting models.

The NSRDB dataset includes both measured and satellite-derived data, ensuring high accuracy and coverage even in regions where ground-based measurements are limited. It captures variations in solar radiation caused by atmospheric conditions such as cloud cover, temperature, and humidity.

The dataset used in this project spans multiple years, allowing the model to learn seasonal patterns, daily cycles, and long-term trends in solar irradiance. This temporal richness is essential for building robust and reliable forecasting models.

Key Features (Attributes)

The dataset consists of several important attributes that directly or indirectly influence solar irradiance levels. These features are used as input variables for training machine learning and deep learning models.

1. Global Horizontal Irradiance (GHI)

GHI represents the total amount of solar radiation received on a horizontal surface. It is the primary target variable in solar forecasting and includes both direct and diffuse radiation components.

2. Direct Normal Irradiance (DNI)

DNI measures the solar radiation received directly from the sun without atmospheric scattering. It is particularly important for concentrated solar power systems.

3. Diffuse Horizontal Irradiance (DHI)

DHI represents the scattered solar radiation reaching the Earth's surface. It helps capture the effect of clouds and atmospheric particles.

4. Temperature

Ambient temperature affects atmospheric conditions and influences solar radiation levels. It is also important for photovoltaic system performance.

5. Relative Humidity

Humidity impacts cloud formation and atmospheric transparency, indirectly affecting solar irradiance.

6. Wind Speed

Wind speed influences temperature distribution and atmospheric conditions, contributing to variations in solar radiation.

7. Solar Zenith Angle

This parameter represents the angle between the sun and the vertical direction. It plays a crucial role in determining the intensity of solar radiation received.

8. Cloud Cover / Sky Condition

Cloud presence significantly affects solar irradiance by blocking or scattering sunlight.

9. Timestamp (Date and Time)

Time-based features such as hour, day, month, and season are essential for capturing periodic patterns in solar radiation.

Data Characteristics

The NSRDB dataset exhibits several important characteristics that make it suitable for solar irradiance forecasting:

1. Time-Series Nature

The dataset is sequential and time-dependent, where each data point is linked to a specific timestamp. This allows models to learn temporal dependencies and trends.

2. High Resolution

Data is available at fine time intervals (30-minute or hourly), enabling detailed analysis of solar radiation patterns throughout the day.

3. Seasonal and Daily Patterns

Solar irradiance follows predictable patterns based on the time of day and season. The dataset captures these variations effectively.

4. Non-Linearity

The relationship between input features and solar irradiance is highly non-linear due to complex atmospheric interactions. This makes advanced models such as LSTM and deep learning approaches more suitable.

5. Missing and Noisy Data

Like most real-world datasets, NSRDB may contain missing values or noise due to measurement errors or environmental factors. Data preprocessing techniques are applied to handle these issues.

6. Large Dataset Size

The dataset contains a significant amount of data spanning multiple years, which is beneficial for training robust machine learning models.

Relevance to Solar Forecasting

The NSRDB dataset is highly relevant for solar irradiance forecasting due to its comprehensive coverage and high-quality data. It provides all the necessary parameters required to model solar energy generation accurately.

Solar forecasting is essential for:

- Renewable energy planning

- Grid stability and load balancing
- Energy storage optimization
- Smart grid management

The dataset captures both direct and indirect factors affecting solar radiation, allowing models to learn complex relationships between environmental conditions and solar output.

Its time-series structure makes it particularly suitable for deep learning models such as Long Short-Term Memory (LSTM) networks, which are designed to handle sequential data.

Usage in This Project

In this project, the NSRDB dataset is used to develop and evaluate machine learning and deep learning models for predicting solar irradiance, specifically Global Horizontal Irradiance (GHI).

The dataset undergoes several preprocessing steps before being used for model training:

- Data cleaning to remove missing or inconsistent values
- Feature selection to identify the most relevant variables
- Normalization to scale data for better model performance
- Time-based feature extraction (hour, day, month)

The processed dataset is then divided into training and testing sets. Multiple models, including traditional machine learning algorithms and deep learning models such as LSTM, are trained on the dataset.

The objective is to predict future solar irradiance values based on historical data and environmental conditions. The performance of each model is evaluated using standard metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 score.

The insights obtained from this analysis help improve forecasting accuracy and support efficient solar energy management.

DATASET:

NSRDB (National Solar Radiation Database) - 1998-2023 Dataset

Dataset Website Link- https://nsrdb.nlr.gov/

Dataset Used Link- https://s3.us-west-2.amazonaws.com/nsrdb-data.stratus.nlr.gov/7cef467de3abbddf649ee4c766c2ddc1.zip